

UEI628: OPTICAL INSTRUMENTATION

L	T	P	Cr
3	0	0	3.0

Course Objectives: To make the students able to understand different aspects of optical instrumentation. To introduce opto-electronic devices and optical components. To explain the concept of interferometry

Light Sourcing, Transmitting and Receiving: Concept of light, classification of different phenomenon based on theories of light, basic light sources and its characterization, polarization, coherent and incoherent sources, grating theory, application of diffraction grating, electro-optic effect, acousto-optic effect and magneto-optic effect.

Opto–Electronic devices and Optical Components: Photo diode, PIN, photo-conductors, solar cells, phototransistors, materials used to fabricate LEDs and lasers design of LED for optical communication, response times of LEDs, LED drive circuitry, lasers classification ruby lasers, neodymium lasers, CO₂ lasers, dye lasers, semiconductors lasers, lasers applications.

Interferometry: Interference effect, radiometry, types of interference phenomenon and its application, michelson’s interferometer and its application refractometer, rayleigh’s interferometers, spectrographs and monochromators, spectrophotometers, calorimeters, medical optical instruments

Optical Fiber Sensors: Active and passive optical fiber sensor, intensity modulated, displacement type sensors, multimode active optical fiber sensor (micro bend sensor) single mode fiber sensor-phase modulates and polarization sensors

Fiber optic fundamentals and Measurements: fundamental of fibers, fiber optic communication system, optical time domain reflectometer (OTDR), time domain dispersion measurement, frequency domain dispersion measurement.

Course Learning Outcomes (CLO): After the completion of the course student will be able to:

5. Exhibit the knowledge of basic concepts of optical transmitting and receiving
6. Exhibit the knowledge of different opto- electronic devices
7. elucidate different methods of interferometry
2. Exhibit the knowledge of selection of the appropriate optical fiber sensors for industrial application
3. Exhibit the knowledge of fibre optic fundamentals

Text books:

1. J.Wilson&J F B Hawkes, *Opto Electronics: An Introduction*, Prentice Hall of India, (2011), 3rd ed.
2. Rajpal S. Sirohi, *Wave Optics and its Application*, (2001), 1st ed.
3. A Yariv, *Optical Electronics*/C.B.S. Collage Publishing, New York, (1985)
4. Pollock, *Fundamentals of Optoelectronics*, (1994)

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30